

Quant Developer

Stefan Seidl, PhD

- Systematic Portfolio Construction
- Mean-Variance Optimization (MVO)
- Disciplined Convex Programming
- Signal Integration & Risk Overlays

Portfolio Variance: The Two-Asset Case

$$\sigma_p^2 = \mathbf{w}^T \Sigma \mathbf{w}$$

$$\sigma_p^2 = \begin{bmatrix} w_A & w_B \end{bmatrix} \begin{bmatrix} \sigma_A^2 & \rho \sigma_A \sigma_B \\ \rho \sigma_B \sigma_A & \sigma_B^2 \end{bmatrix} \begin{bmatrix} w_A \\ w_B \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \text{Var}_A & \text{Cov}_{AB} \\ \text{Cov}_{AB} & \text{Var}_B \end{bmatrix}$$

- $\mathbf{w}$: Portfolio weights $\longrightarrow \mathbf{1}^T \mathbf{w} = 1$
- Σ : **Covariance matrix** (positive semi-definite)
- $\rho \in [-1, 1]$: Correlation $\longrightarrow$ reflect **actual risk** in portfolio
- $\sigma_i^2 > 0$: Single variance ($\sigma_i^2 = 0 \iff$ const. return, e.g. fixed rates)

Portfolio Optimization is Convex

Conditions for a Global Minimum:

- **First Order:** $\nabla f(\mathbf{w}) = 0$ (Stationary point)
 - **Second Order:** $\nabla^2 f(\mathbf{w}) = \mathbf{H} \succeq 0$ (Positive semi-definite Hessian)
-

Proof: Positive Semi-Definiteness (PSD) and Eigenvalues

Let $\mathbf{v}$ be an eigenvector of Σ with eigenvalue λ :

$$\Sigma \mathbf{v} = \lambda \mathbf{v}$$

$$\mathbf{v}^T \Sigma \mathbf{v} = \mathbf{v}^T (\lambda \mathbf{v}) = \lambda (\mathbf{v}^T \mathbf{v}) = \lambda \|\mathbf{v}\|^2$$

Since $\|\mathbf{v}\|^2 \geq 0$, the quadratic form $\mathbf{v}^T \Sigma \mathbf{v} \geq 0$ if and only if $\lambda \geq 0$.

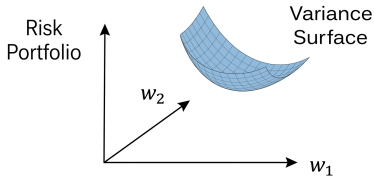
Summary for Portfolio Context:

- $\Sigma \succeq 0$ (PSD) $\iff$ All eigenvalues $\lambda_i \geq 0$
 - Variance $f(\mathbf{w}) = \mathbf{w}^T \Sigma \mathbf{w}$ is **convex** $\iff \Sigma$ is PSD
-

Note: This property ensures that any local minimum is also the global minimum.

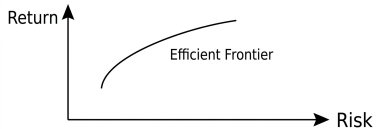
Visualizing Convex Portfolio Optimization

Variance Surface



- Quadratic form $\mathbf{w}^T \Sigma \mathbf{w}$
- Convex shape (bowl)
- Global minimum exists

Efficient Frontier (EF)



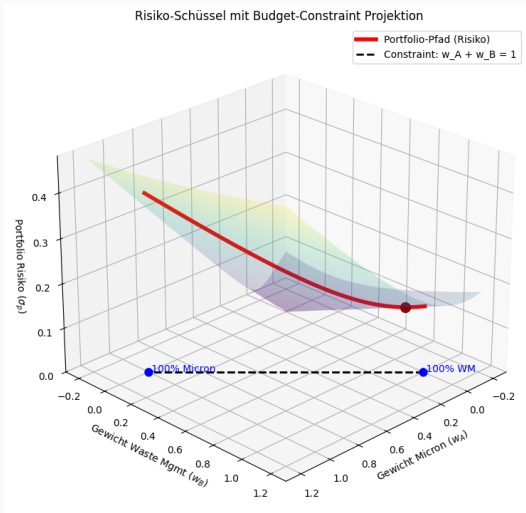
- Risk vs. Return trade-off
- Set of optimal portfolios
- Solved via convex optimization

EF: Corresponds to a cross-section of the variance surface

EF: Sequence of points providing maximum return for each risk level

Stefan: “Schnitt durch die Risiko-Schüssel”

Visualization of the Variance Surface

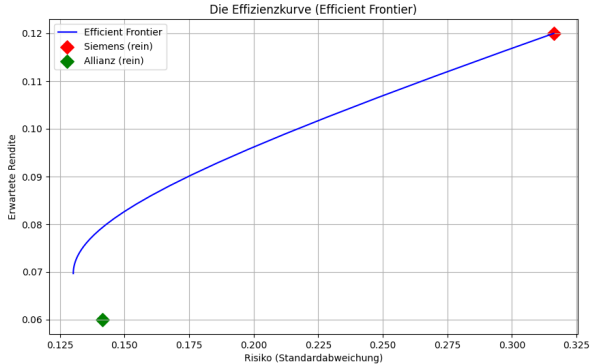


- $1 = w_{MU} + w_{WM}$
- Risk increases with higher **Micron** exposure
- Low risk due to significant **Waste Management** allocation

The convex shape guarantees a unique Minimum Variance Portfolio.

Note: Variance surface calculated via Python (CVXPY/NumPy); returns based on simulated price data.

Efficient Frontier



- **Left:** Minimum Variance Portfolio (MVP)
- **Right:** Maximum Return Portfolio
- **Below:** Inefficient portfolios (suboptimal)
- **Above:** Mathematically impossible region

Minimizing Portfolio Variance

$$\min_w \frac{1}{2} \sigma^2 = \min_w \frac{1}{2} \mathbf{w}^T \Sigma \mathbf{w} \quad (\text{Objective Function})$$

For two assets A and B , the variance is:

$$\sigma^2 = \sigma_A^2 w_A^2 + \sigma_B^2 w_B^2 + 2w_A w_B \rho \sigma_A \sigma_B$$

Definitions:

- $\rho = \frac{\text{Cov}(A, B)}{\sigma_A \sigma_B}$ (Normalization)
- $\text{Cov}(A, B) = \frac{1}{n-1} \sum_{i=1}^n (R_{A_i} - \bar{R}_A)(R_{B_i} - \bar{R}_B)$

Optimization (Lagrange):

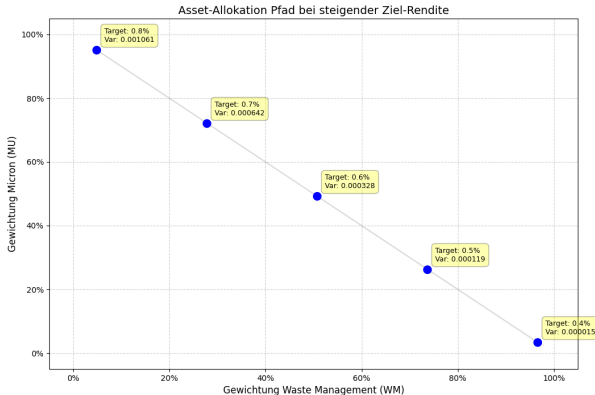
- $\frac{d\sigma^2}{dw_A} \stackrel{!}{=} 0$
- Subject to: $1 = w_A + w_B$

$$w_A = \frac{\sigma_B^2 - \text{Cov}(A, B)}{\sigma_A^2 + \sigma_B^2 - 2\text{Cov}(A, B)}$$

Interpretation:

1. If σ_B^2 is large $\rightarrow$ Numerator increases $\rightarrow$ Flight to asset A
2. If $\text{Cov}(A, B)$ becomes negative $\rightarrow$ Denominator increases $\rightarrow$ Diversification effect

Customer Case: Increasing Target Return



- **Shift in weights** due to higher target return
- Consequently portfolio **variance increases** by moving along EF

The optimizer requires a lower bound for returns: $\mathbf{w}^T \boldsymbol{\mu} \geq \mu_{\text{Target}}$

Relative Strength Index (RSI) - Momentum Indicator

```
stefan@DESKTOP-DUF100G: ~ % python RSI_PayPal.py
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
      Kurs_USD  FX_Rate  Kurs_EUR   RSI  Signal
Date
2026-01-14    57.49    1.16    49.38  43.11  HOLD
2026-01-15    56.57    1.16    48.57  35.31  HOLD
2026-01-16    56.72    1.16    48.86  37.79  HOLD
2026-01-20    54.91    1.16    47.18  32.22  HOLD
2026-01-21    55.72    1.17    47.51  35.90  HOLD
2026-01-22    56.98    1.17    48.81  46.25  HOLD
2026-01-23    56.45    1.18    48.02  43.70  HOLD
2026-01-26    56.43    1.19    47.59  34.86  HOLD
2026-01-27    55.34    1.19    46.60  28.96  BUY
2026-01-28    53.92    1.20    44.86  27.00  BUY
2026-01-29    52.95    1.20    44.21  25.82  BUY
2026-01-30    52.53    1.20    43.90  26.12  BUY
2026-02-02    52.17    1.18    44.03  27.37  BUY
2026-02-03    41.57    1.18    35.24  16.03  BUY
2026-02-04    40.91    1.18    34.62  10.85  BUY
2026-02-05    39.78    1.18    33.71  10.79  BUY
2026-02-06    40.30    1.18    34.22  11.83  BUY
2026-02-09    41.03    1.18    34.69  15.27  BUY
2026-02-10    41.36    1.19    34.75  13.95  BUY
2026-02-11    40.34    1.19    33.93   6.80  BUY
2026-02-12    38.96    1.19    32.81   6.67  BUY
2026-02-13    40.17    1.19    33.84  12.16  BUY
2026-02-17    40.69    1.18    34.34  15.28  BUY
2026-02-18    41.32    1.19    34.87  19.63  BUY
2026-02-19    41.60    1.18    35.29  22.52  BUY
2026-02-20    41.52    1.18    35.28  22.94  BUY
2026-02-23    43.92    1.18    37.11  30.36  HOLD
2026-02-24    46.88    1.18    39.75  69.67  HOLD
2026-02-25    47.18    1.18    40.06  74.38  SELL
```

$$RSI = 100 - \frac{100}{1 + RS}$$

$$RS = \frac{\frac{1}{n} \sum \text{Gains}}{\frac{1}{n} \sum |\text{Losses}|}$$

- Momentum indicators aim to predict **trend reversals**.
- A falling price often indicates deteriorating **expected returns** or shifting **correlations** within the portfolio.

⇒ Identifying oversold (< 30) and overbought (> 70) **not sufficient** for rebalancing

⇒ RSI ignores the portfolio's overall convexity

Convex vs. Concave Optimization - Two Equivalent Ways

Risk Minimization

Focus on the **Convex Bowl**:

$$\min \frac{1}{2} \mathbf{w}^T \Sigma \mathbf{w}$$

cp.**Minimize**(risk)

Utility Maximization

Focus on the **Concave Hill**:

$$\max \left(-\frac{1}{2} \mathbf{w}^T \Sigma \mathbf{w} \right)$$

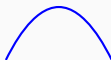
cp.**Maximize**(-risk)

DCP Rules (Disciplined Convex Programming):

- ⇒ **CVXPY** only allows cp.Maximize() for **concave** functions
- ⇒ Since risk is convex, negative risk (later utility) is concave
- ⇒ Both approaches yield the **exact same** optimal weights w



Convex (Min)



Concave (Max)

Objective: Utility Maximization with linear Risk and Tax

$$\max \left(\mu^T \mathbf{w} - \frac{\lambda}{2} \mathbf{w}^T \Sigma \mathbf{w} - \text{tax}^T \mathbf{w} \right)$$

```
w = cp.Variable(n)

p_return      = expected_returns @ w
p_risk        = cp.quad_form(w, cov_matrix)
tax_penalty   = tax_impact @ w

utility = p_return - (lambda_risk / 2) * p_risk - tax_penalty

objective = cp.Maximize(utility)

constraints = [cp.sum(w) == 1, w >= 0, w <= 0.1 ]

prob = cp.Problem(objective, constraints)
prob.solve()
```

Summary & Outlook - Beyond Markowitz

- **Backward-looking:** Classic MPT relies solely on **historical data**
- **Instability:** Small return changes can lead to massive weight shifts
- **Missing Fundamentals:** Model ignores company's health

Thinking Out of the Box: Fund. Integration of **Equity Ratio**

Idea 1: Fundamental Covariance Σ_{Equity} (Risk Penalty):

$$U = \mu^T \mathbf{w} - \underbrace{\frac{\lambda}{2} \mathbf{w}^T \Sigma_{Return} \mathbf{w}}_{\text{Market Risk}} - \underbrace{\frac{\delta}{2} \mathbf{w}^T \Sigma_{Equity} \mathbf{w}}_{\text{Fundamental Risk}} - \text{Tax}^T \mathbf{w}$$

Idea 2: Equity Ratio as Linear Benefit:

$$U = \mu^T \mathbf{w} - \frac{\lambda}{2} \mathbf{w}^T \Sigma_{Return} \mathbf{w} + \gamma \text{Equity}^T \mathbf{w} - \text{Tax}^T \mathbf{w}$$

Idea 3: Equity Ratio as Constraint (e.g., $\text{Equity}^T \mathbf{w} \geq \text{threshold}$)

- + **Stability:** Fundamental ratios are far **less volatile**
- + **Resilience:** Provides a buffer against **shifting environments**